together with the application for type approval.
3.2.2 Test specification details
3.2.2.1 Equipment should operate satisfactorily on completion of each of the

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of this annex, should be submitted to the Administration by the manufacturer,
together with the application for type approval.
3.2.2 Test specification details
3.2.2.1 Equipment should operate satisfactorily on completion of each of the
following environmental tests:
.1 Vibration tests
. 1.1 a search should be made for resonance over the following range
of frequency and amplitude of acceleration:
.1.1.1
1,1.2
. 1.2
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2 to 13.2 Hz with an amplitude of ± 1 mm; and
13.2 to 80 Hz with an acceleration of ± 0.7 g .
This search should be made in each of the three planes at a
rate sufficiently low to permit detection of resonance;
the equipment should be vibrated in the planes at each major
resonant frequency for a period of two hours;
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.1.3 if there is no resonant frequency, the equipment should be
vibrated in each of the planes at 30 Hz with an acceleration
of ± 0.7 g for a period of two hours;
. 1.4 after completion of the tests specified in .1.2 or .1.3 of

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.1.3 if there is no resonant frequency, the equipment should be
vibrated in each of the planes at 30 Hz with an acceleration
of ± 0.7 g for a period of two hours;
. 1.4 after completion of the tests specified in .1.2 or .1.3 of
this paragraph a search shou+d again be made for resonance and
there should be no significant change in the vibration pattern
2 Temperature tests
.2 . 1 equipment that may be installed in an enclosed spaqe that is
environmentally controlled, including an engine room, should
be subjected, for a period of not less than two hours, to:
.2.1.2
a low temperature test at O°C; and
a high temperature test at 55°C.
At the end of each of the tests referred to , the equipment

should be switched on and it should function normally under the test conditions .

3 Humidity tests

. 3 . 1 equipment should be left switched off for a period of two hours at a temperature of 55°C in an atmosphere with a relative humidity of 90~. At the end of this period the equipment should be switched on and should operate sat~sfactorily for one hour

4 Inclination test

. 4.1 equipment should operate satisfactorily at angles of

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relative humidity of 90~. At the end of this period the equipment should be switched on and should operate sat~sfactorily for one hour

4 Inclination test

. 4.1 equipment should operate satisfactorily at angles of inclination up to 22.5° in any plane from the normal operating position

5 Reliability of electrical and electronic equipment

.5.1 the electrical and electronic components of the equipment should be of a quality g~aranteed by the manufacturer and suitable for their intended purpose.

PART 4 - METHOD FOR DETERMINATION OF OIL CONTENT

4.1 Scope and application

4.1.1 The method includes the measurement of most light oil fractions, although some loss of volatile components will occur during the extractions.

4.1.2 The method has a nominal working range from 2 to 80 mg/l. The lower level of detection can be improved to 0.1 mg/l by using longer path-length cells. The upper limit of the method ca~ be ex~ended at least to 1,000 mg/l by preparing dilutions of the sample extract.

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by preparing dilutions of the sample extract.

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4.2 Summary of method

The sample is acidified to a low pH and extracted with two volumes of carbon tetrachloride. The oil content is determined by comparison of the infrared absorbances of the sample extract against known concentrations of the appropriate reference oil. Other suitable non-infrared active solvents may be used if perferred.

4.3 Sample and storage

4.3.1 A representative sample of 1 litre
glass bottle with a pressure-sealing tap.
on the day of collection, it is preserved
hydrochloric acid (HCl) (4.5.1).
volume is colleced in a narro~-neck
Unless the sample will be extracted
with the addition of 5 ml

4.3.2 Because losses of oily matter will occur on samplirig equipment, the collection of a composite sample is impractical~ Individual portions collected at prescribed time intervals must be analysed separately to obtain the average concentration over an extended period.

4.4 Apparatus

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collection of a composite sample is impractical~ Individual portions collected at prescribed time intervals must be analysed separately to obtain the average concentration over an extended period.

4.4 Apparatus

4.4.1 Separatory funnel, 1,000 ml volume, with Teflon stopcock.

4.4.2 Infrared spectrophotometer.

4.4.3 Cells, 5 mm path-length, sodium chloride or infrared-grade quartz with a minimum of 80~ transmittance at 2,930 cm- I The 5 mm path-length is recommended as being convenient for monitoring levels normally encountered. Longer path-lengths may be used.

4.4.4 Filter paper, medium grade, 12 '.5 cm.

4.5 Reagents

4.5.1 Hydrochloric acid, HCl 1:1. Mix equal amounts of concentrated HCl and distilled water.

4.5.2 Sodium chloride, NaCl reagent grade.

4.5.3 Carbon tetrachloride, CC14 reagent grade~

4.5.4 Oil reference: Oil collected from the source at the same time the sample was collected.

4.5.5 Stock reference standard (3 mg/ml): accurately'weigh about 0.30g of reference oil (4.5.4) into a tared 100 ml volumetric flask and dilute to volume with carbon tetrachloride. Using the reference oil at room ambient

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sample was collected.

4.5.5 Stock reference standard (3 mg/ml): accurately'weigh about 0.30g of reference oil (4.5.4) into a tared 100 ml volumetric flask and dilute to volume with carbon tetrachloride. Using the reference oil at room ambient
temperature, record the relative density and temperature of the reference oil. A weight-to-volume conversion to volume-to-volume must take into account the differing densities of the two liquids used in preparing the calibration

plot (4.8.2).
4.5.6 Calibration standards: prepare a series of dilutions by pipetting volumes of stock reference standards into 100 ml volumetric flasks and
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diluting to volume with carbon tetrachloride. A convenient series of volumes is 5, 10, 15, 20, and 25 ml of stock solution. Calculate the exact concentrations of the dilution in millilitres times ·ten to the minus three per hundred millilitres of solvent (ml x 10- 3 /100 ml) from the information above (4.5.5).
4.6 Extraction

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concentrations of the dilution in millilitres times ·ten to the minus three per hundred millilitres of solvent (ml x 10- 3 /100 ml) from the information above (4.5.5).